



Science — Cbse Annual Exam — 80 Marks

Subject: Science | Class: Class 8

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	How do particles behave in materials that are gases compared to solids? (A) They cannot move much and are fixed in position (B) They are bonded together to form pure substances (C) They combine physically to form mixtures only (D) They can move around freely <i>[Ch 1: Exploring the Investigative World of Science]</i>	1
2	What does the image of a root on the left-hand pages of the textbook symbolize? (A) The need to fly high in scientific exploration (B) The starting point of all experiments (C) The deep, solid foundation of knowledge connected to environment, traditions, and heritage (D) The importance of plants in science <i>[Ch 1: Exploring the Investigative World of Science]</i>	1
3	Assertion (A): Human-made ecosystems like agricultural fields must be made sustainable. Reason (R): Agricultural fields are necessary ecosystems. (A) A is true but R is false. (B) Both A and R are true, and R is the correct explanation of A. (C) Both A and R are true, but R is NOT the correct explanation of A. (D) A is false but R is true. <i>[Ch 12: How Nature Works in Harmony]</i>	1



Q.NO.	QUESTIONS	MARKS
4	Which of the following forms a non-uniform mixture when mixed with water? (A) Chalk powder (B) Sugar (C) Salt (D) Oxygen gas <i>[Ch 9: The Amazing World of Solutes, Solvents, and Solutions]</i>	1
5	When a bicycle stops pedalling on a flat road, the bicycle slows down and comes to a stop. What causes this change in speed? (A) A force is acting on the bicycle to bring it to rest (B) No force acts on the bicycle (C) The gravitational force pulls it backward (D) The muscular force of the rider continues to act <i>[Ch 5: Exploring Forces]</i>	1
6	When the ends of the cylindrical coil are connected to the terminals of an electric cell, what happens to the magnetic compass needles placed near the ends of the coil? (A) They remain completely stationary (B) They point vertically upward (C) They show deflection from their original positions (D) They lose their magnetic properties <i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	1
7	What does the syringe experiment demonstrate about gas particles when the plunger is pushed inward? (A) Gas particles have space between them that can be reduced by applying external pressure (B) Gas particles disappear when compressed (C) Gas particles change into liquid particles when compressed (D) Gas particles have no space between them <i>[Ch 7: Particulate Nature of Matter]</i>	1
8	Dr. Ananda Mohan Chakrabarty developed a bacterium in 1971 that could: (A) Decompose plastic waste (B) Break down oil spills (C) Ferment flour for cooking (D) Produce biogas from household waste <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	1



Q.NO.	QUESTIONS	MARKS
9	Why are overhead water tanks placed at a height in buildings? (A) To make them easier to fill with water (B) To increase pressure in the taps, resulting in a good stream of water (C) To protect them from damage at ground level (D) To reduce the cost of construction (E) To reduce evaporation of stored water <i>[Ch 6: Pressure, Winds, Storms, and Cyclones]</i>	1
10	By how much does Makar Sankranti move ahead in the tropical calendar? (A) One day every 50 years (B) One day every 71 years (C) One day every year (D) One day every 100 years <i>[Ch 11: Keeping Time with the Skies]</i>	1
11	Assertion (A): The range of distances from a star where water remains liquid is called the habitable zone. Reason (R): The habitable zone is also known as the Goldilocks zone. (A) A is false but R is true. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) Both A and R are true, and R is the correct explanation of A. <i>[Ch 13: Our Home: Earth, a Unique Life Sustaining Planet]</i>	1
12	The human eye contains which type of lens inside it? (A) Concave lens (B) Plane mirror (C) Convex lens (D) Flat glass plate <i>[Ch 10: Light: Mirrors and Lenses]</i>	1
13	Assertion (A): The Moon takes about 50 minutes longer than the previous day to appear in nearly the same position in the sky. Reason (R): While the Earth completes one rotation in 24 hours, the Moon moves ahead in its orbit around the Earth during this time. (A) A is true but R is false. (B) A is false but R is true. (C) Both A and R are true, and R is the correct explanation of A. (D) Both A and R are true, but R is NOT the correct explanation of A. <i>[Ch 11: Keeping Time with the Skies]</i>	1



Q.NO.	QUESTIONS	MARKS
14	Why are spices with salt or sugar added to food items like pickles? (A) They act as preservatives and do not allow microorganisms to grow (B) They reduce the nutritional value of food (C) They improve the color of the food (D) They make the food easier to cook <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	1
15	Assertion (A): Human beings are not crushed under the atmospheric pressure. Reason (R): The pressure inside our bodies is equal to the atmospheric pressure, which balances the pressure from outside. (A) Both A and R are true, but R is NOT the correct explanation of A. (B) Both A and R are true, and R is the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 6: Pressure, Winds, Storms, and Cyclones]</i>	1
16	If Earth were much smaller than its actual size but had the same average density, what would happen to its atmosphere according to the chapter? (A) The atmosphere would have more oxygen for breathing (B) The gases in the atmosphere would escape into space due to weak gravity (C) The atmosphere would become thicker and denser (D) The atmosphere would remain exactly the same <i>[Ch 13: Our Home: Earth, a Unique Life Sustaining Planet]</i>	1
17	Assertion (A): Hydrogen and oxygen present in water cannot be separated by physical methods, but sodium chloride dissolved in water can be separated by the physical process of evaporation. Reason (R): The constituent elements of a compound cannot be separated by any physical method, but components of a mixture can be separated by physical processes. (A) A is false but R is true. (B) Both A and R are true, and R is the correct explanation of A. (C) A is true but R is false. (D) Both A and R are true, but R is NOT the correct explanation of A. <i>[Ch 8: Nature of Matter: Elements, Compounds, and Mixtures]</i>	1
18	Why are there fewer dragonflies in ponds with fish compared to ponds without fish? (A) Fish eat dragonfly larvae (B) Dragonflies cannot fly near fish (C) Fish compete with dragonflies for food (D) Dragonflies avoid ponds with water <i>[Ch 12: How Nature Works in Harmony]</i>	1



Q.NO.	QUESTIONS	MARKS
19	In traditional Indian practice, variolation was used to protect against smallpox. What did this method involve? (A) Administering an inactivated bacterial toxin (B) Using material from a smallpox sore to scratch the skin (C) Injecting cowpox sap into a healthy person (D) Taking medicines called antibiotics <i>[Ch 3: Health: The Ultimate Treasure]</i>	1
20	Sample B (iron sulfide) is NOT attracted by a magnet because: (A) Iron sulfide contains no iron (B) Iron sulfide is too heavy to be attracted (C) Iron sulfide is a compound with completely different properties than its constituents (D) Iron sulfide is a liquid <i>[Ch 8: Nature of Matter: Elements, Compounds, and Mixtures]</i>	1
Section B Attempt 5 out of 7.		
21	How does pressure difference affect air movement? Give examples of weather phenomena that result from different pressure differences. <i>[Ch 1: Exploring the Investigative World of Science]</i>	2
	OR State the location and function of the microorganism associated with pulse crops. <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	
22	How is gravitational force different from magnetic force and electrostatic force in terms of being attractive or repulsive? <i>[Ch 5: Exploring Forces]</i>	2
	OR State two reasons why nichrome wire is commonly used in electrical heating devices. <i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	
23	Why is it advised to keep doors and windows open during storms with high-speed winds? <i>[Ch 6: Pressure, Winds, Storms, and Cyclones]</i>	2
24	Fill in the blanks: (i) The solution used in a Voltaic cell is called ____. (ii) A current carrying coil behaves like a ____. <i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	2
25	Why do antibiotics not work against viral infections?	2



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 3: Health: The Ultimate Treasure]</i>	
Section C Attempt 5 out of 7.		
26	What is antibiotic resistance? Explain how it develops in bacteria and list three precautions to prevent its spread. <i>[Ch 3: Health: The Ultimate Treasure]</i>	3
	OR Explain the basic structure of a cell and describe the functions of the cell membrane and cytoplasm. <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	
27	The weight of an object on the Moon becomes one-sixth of its weight on the Earth. What causes this change? Does the mass of the object also become one-sixth of its mass on the Earth? <i>[Ch 5: Exploring Forces]</i>	3
	OR What are interparticle attractions? Explain how the strength of these attractions affects the arrangement and movement of particles in solids. <i>[Ch 7: Particulate Nature of Matter]</i>	
28	A boy lies horizontally on a loose sand bed. The same boy then stands vertically on the same sand bed. In which case will he sink more? Explain your answer with proper reasoning. <i>[Ch 6: Pressure, Winds, Storms, and Cyclones]</i>	3
29	How do pressure differences lead to different weather conditions? Explain with examples. <i>[Ch 1: Exploring the Investigative World of Science]</i>	3
30	How does inserting an iron nail inside the paper cylinder affect the magnetic effect produced by a current-carrying coil, and what observation confirms this? <i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	3
Section D Attempt 4 out of 6.		
31	What is pressure? Explain with a numerical example how pressure is calculated. How does the area over which a force is applied affect the pressure? Give two examples from daily life to support your answer. Also explain with an activity how liquids exert pressure. <i>[Ch 6: Pressure, Winds, Storms, and Cyclones]</i>	5



Q.NO.	QUESTIONS	MARKS
	OR Explain the process of investigating a phenomenon like a scientist. Using the example of a puri puffing up in hot oil, describe the key steps involved and list the factors that can be changed or controlled during such an investigation. <i>[Ch 1: Exploring the Investigative World of Science]</i>	
32	Forces can produce different effects on objects. What are the various effects that a force can have on the bodies on which it is applied? Explain each effect with a suitable example and classify the action as push or pull. <i>[Ch 5: Exploring Forces]</i>	5
	OR Describe an activity using a coil, an electric cell, and magnetic compasses to demonstrate how electric current affects magnetic materials. Explain the observations when the current is switched on and off, and how inserting an iron nail inside the coil changes the results. <i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	
33	Explain what non-communicable diseases are with suitable examples. How do lifestyle-related diseases develop, and what steps can help prevent them? Also, describe the role of immunity in protecting the body from diseases. <i>[Ch 3: Health: The Ultimate Treasure]</i>	5
34	Describe the step-by-step procedure for preparing an onion peel slide for microscopic observation. Explain the purpose of each material used in the process and state what observations you would make when viewing the slide under a microscope. <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	5
Section E Attempt 3 out of 4.		
35	Case Study: Effects of Electric Current Ravi conducts an experiment for his science exhibition. He connects a nichrome wire to a cell and observes that the wire becomes warm after some time. He places a magnetic compass near the wire and notices the needle deflects from its original position. He then constructs an electromagnet by wrapping nichrome wire around an iron nail. He prepares three electromagnets with 25, 50, and 100 turns respectively to compare their strength. He also tests coils made of copper and aluminium wire to observe compass deflection. (i) Name the two effects of electric current demonstrated in Ravi's experiment. (ii) State one reason why nichrome is preferred over copper for making heating elements in electric devices. (iii) Predict how the lifting capacity of the electromagnet changes when the number of turns increases from 25 to 100. Explain your answer. (iv) Will the compass needle show deflection when placed near a coil made of aluminium wire carrying current? Justify your answer.	5



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	
	OR Case Study: Systematic Scientific Investigations Class 8 students are conducting systematic investigations to understand everyday phenomena. One group is studying how puris puff up by changing only the oil temperature while keeping dough thickness and dropping method identical. They record yes/no observations about puffing and measure the time in seconds it takes to puff. Another group examines how we use electric current in daily life—depending on its heating effect to keep warm and magnetic effect to run motors. A third group investigates health by exploring how tiny unseen microbes include both helpful ones that aid digestion and harmful ones that cause infections, and how medicines and vaccines help us stay healthy. (i) What specific numerical measurement does the first group record when studying how puris puff up? (ii) Why does the group use circles of dough of the same thickness and drop them in the same way when testing different oil temperatures? (iii) Identify the two effects of electric current mentioned in the investigation and state one practical use of each effect. (iv) Distinguish between helpful microbes and harmful microbes based on their effects on the human body. <i>[Ch 1: Exploring the Investigative World of Science]</i>	
36	Case Study: Exploring the World of Microorganisms Riya, a Class 8 student, conducted experiments to study microorganisms. She collected pond water and observed it under a microscope that magnifies objects 100 to 400 times. She recorded seeing Amoeba with an irregular shape and Paramecium that moved using specialised structures. She also prepared a soil suspension from garden soil and observed tiny organisms in it. Later, she noticed a cotton-like growth on a tomato left outside for several days. When she asked why homemade pickles never showed such growth, her grandmother explained that adding salt and spices acts as preservatives. (i) Define microorganisms. (ii) Why is a microscope necessary to observe the organisms in pond water? (iii) Based on Riya's observations, identify the category of microorganisms for Amoeba and Paramecium. Write one characteristic of each that she recorded. (iv) Explain why pickles preserved with salt and spices do not get infected by microorganisms easily. <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	5



Q.NO.	QUESTIONS	MARKS
37	Case Study: Health and Disease Management During Health Week, students learned that health means complete physical, mental, and social well-being—not just the absence of disease. The doctor explained that infectious diseases are caused by pathogens like bacteria and viruses, while non-communicable diseases like diabetes result from lifestyle factors. She clarified that symptoms are what patients feel, whereas signs are visible or measurable. The program highlighted how vaccines train the immune system using dead or weakened germs, and emphasized that early diagnosis with continuous care helps control non-communicable diseases through lifestyle changes and medication. (i) What is the complete definition of health? (ii) Differentiate between 'symptoms' and 'signs' of a disease. (iii) Suggest two strategies for managing non-communicable diseases like diabetes to improve quality of life. (iv) Explain why vaccines are effective against infectious diseases but not against diabetes. <i>[Ch 3: Health: The Ultimate Treasure]</i>	5

